

Dr. Thejus Mahajan

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Hamburg, Germany

CAREER INTERESTS To apply expertise in computational modeling, data science, and bioinformatics to address complex challenges in clinical and biological research. Skilled in multivariate statistical modeling, biostatistics, and developing analysis pipelines using R (Bioconductor, Tidymodels) and Python. Actively transitioning into clinical bioinformatics through specialized training on patient datasets and biostatistical methods for health research.

- WORK EXPERIENCE
- **CQ Beratung + Bildung, Berlin, Germany**
Further Training – Application-related Bioinformatics and Biostatistics
August 2025 – Present
 - Shell (BASH), Python (object-oriented), and programming methodology.
 - Bioinformatics databases (NCBI, Biopython, SQL).
 - Sequence analysis, structural bioinformatics, and NGS analysis (Nextflow, Galaxy).
 - Applied biostatistics with real and simulated patient datasets (R/Bioconductor, ANOVA, PCA, HCA, survival analysis).
 - **HealthTwist GmbH, Berlin, Germany**
Internship – Machine Learning in R
February 2026 – April 2026
 - Implementation of Machine Learning workflows using the Tidymodels framework in R.
 - Supervised by Dr. Andreas Busjahn (HealthTwist GmbH).
 - Practical application of bioinformatics and biostatistical analysis skills.
 - **Helmholtz-Zentrum Hereon, Geesthacht, Germany**
Guest Scientist – Computational Biologist & Systems Modeler
May 2025 – October 2025
 - Spearheaded research modeling evolutionary processes in marine environments.
 - Simulated long-term impacts of environmental changes on ecosystems.
 - **University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences**
Post-doctoral Scientist – Marine Ecosystem Modelling
August 2021 – January 2025
 - Developed the “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework.
 - Engineered models in Fortran and analyzed large NetCDF datasets using Python in Linux.
 - Statistical analysis and visualization of ecological time-series data using R.

	• Université Paris-Saclay, Institut des Sciences Moléculaires d'Orsay, France Ph.D. Researcher in Astrochemistry October 2015 – September 2018 - Led atomic collision experiments and processed terabytes of raw data. - Utilized C++ for signal processing and algorithm development. - Developed a molecular fragmentation model for the KIDA database. - Thesis: “Excitation and fragmentation of C_nN^+ ($n=1-3$) molecules in collisions with He atoms at intermediate velocity; fundamental aspects and application to astrochemistry.” - Advisor: Dr. Karine Béroff	
EDUCATION	• Ph.D. in Astrochemistry (2015 – 2018) Institut des Sciences Moléculaires (ISMO), Université Paris-Saclay, France • M.Sc. in Physics (July 2012 – January 2015) National Institute of Technology Calicut, India CGPA: 8.2/10	
SKILLS & COMPETENCIES	Programming: Bioinformatics: Biostatistics: Data Analysis: Tools:	Python (Pandas, NumPy, Scikit-learn), R (Bioconductor, Tidy-models, ggplot2), Fortran, SQL, BASH NGS Analysis, Galaxy, Sequence Alignment, Structural Bioinformatics, Nextflow Multivariate Analysis, PCA, ANOVA, Dimensionality Reduction, Markov Chains Large dataset processing (NetCDF, HDF5), Data visualization, Pipeline development Linux/Unix, Git, HPC clusters, $\text{\LaTeX}$, Microsoft Office
LANGUAGES	English: German: French: Malayalam: Tamil, Hindi:	Proficient (speaking, reading, writing) B1 (Goethe Certified) Intermediate Native Proficient
TRAINING & COURSES	- HPC Training at Jülich Supercomputing Center (JSC) - MITx: 7.00x Introduction to Biology – The Secret of Life	
PUBLICATIONS	Google Scholar: https://scholar.google.com/citations?hl=en&user=PJkZwAwAAAAJ 1. T. Mahajan, et al. <i>Journal of Physics: Conference Series</i>, 1412:142026, 2020. 2. T. Idbarkach, et al. <i>Astronomy & Astrophysics</i>, 628:A75, 2019. 3. T. Mahajan, et al. <i>Journal of Physics B</i>, 2019. 4. T. Idbarkach, et al. <i>Journal of Physics B</i>, 51(24):245201, 2018. 5. T. Idbarkach, et al. <i>Journal of Physics: Conference Series</i>, 1412(11):112028, 2020.	

REFERENCES

- **Dr. Andreas Busjahn**
HealthTwist GmbH, Berlin
Email: andreas@busjahn.net
Professional relation: Biostatistics instructor and internship supervisor
- **Prof. Dr. Kai Wirtz**
Helmholtz-Zentrum Hereon, Ökosystemmodellierung
Email: kai.wirtz@hereon.de
Professional relation: Supervisor during Guest Scientist position
- **Prof. Dr. Elisa Schaum**
University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences
Email: elisa.schaum@uni-hamburg.de
Professional relation: Collaborator during Post-doctoral research